1. Genetic Algorithm

Given:

* Function: f(x)=x^2
* Initial population = {2, 4, 6, 8}

Step 1: Calculate Fitness Values

Individual (x)	Fitness = x^2
2	4
4	16
6	39
8	64

The individuals with the highest fitness are:

* 8 (Fitness = 64)
* 6 (Fitness = 36)

These are selected as the best candidates.

Step 2: Binary Representation

Number	binary
6	0110
8	1000

Step 3: Crossover (Exchange Least Significant Bits)

The least significant bit (LSB) is the rightmost bit.

* 0110 (6) → LSB = 0
* 1000 (8) → LSB = 0

After exchanging the LSBs, the values remain:

* 0110 = 6
* 1000 = 8

Offspring produced: 6 and 8.

(If the LSBs were different, the offspring would have changed.)

Step 4: Role of Mutation

Mutation:

* Randomly changes one or more bits.
* Increases population diversity.
* Prevents the algorithm from getting stuck at local optimum solutions.
* Helps discover better solutions over generations.

Answer:

* Fitness values = {4, 16, 36, 64}
* Best candidates = 6 and 8.
* Offspring after crossover = 6 and 8.
* Mutation improves diversity and the effectiveness of the Genetic Algorithm.

2. Perceptron Problem

Given:

* Advertisement exposure (x_1) = 2
* Income level (x_2) = 1
* Weight (w_1) = 0.5
* Weight (w_2) = 0.3
* Bias (b) = –0.4

Step 1: Calculate Net Input

Formula:

Net Input = (w_1 × x_1) + (w_2 × x_2) + b

Substitute the values:

= (0.5 × 2) + (0.3 × 1) + (−0.4)

= 1 + 0.3 − 0.4

= 0.9

Step 2: Apply Step Activation Function

The step function is:

* If Net Input ≥ 0 → Output = 1
* If Net Input < 0 → Output = 0

Since:

$0.9 > 0$

The output is:

1

Step 3: Interpretation

* Output = 1 → Customer will purchase the product.

Parameter	Value
Net input	0.9
Perception output	1
Decision	Customer will purchase the product